# CHARLOTTE DEMAS

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## EDUCATION

**Wake Forest University** **Winston-Salem, NC**

*B.A. Economics & B.S. Computer Science* *Class of 2025*

*GPA: 3.6*

DIS Copenhagen, Study Abroad 2024: Artificial Neural Networks & Deep Learning

## PROFESSIONAL EXPERIENCE

***Software Engineer I - Application and Database Services*** **Jersey City, NJ**

*Bank of America* *July 2025 - present*

- Drove a proof of concept deploying IBM Application Modernization Accelerator on OpenShift, securing restricted namespace provisioning, cluster-policy exceptions and privileged access to pave the way for app teams across the bank to migrate from VMs to containers
- Automate provisioning and configuration of Java application and web servers across AWS and private cloud with Terraform and Ansible, cutting environment setup from 2+ hours to under 15 minutes
- Own incident response for 5 production applications; resolved 400+ incidents and reduced repeat issues by 35%
- Migrated middleware provisioning from a legacy internal tool to Terraform Enterprise, consolidating 3 products onto a single provisioning platform for all app teams

***Global Technology Summer Analyst*** **Jersey City, NJ**

*Bank of America* *June 2024 - August 2024*

- Built a Python backend for an internal hackathon that summarizes unread email and ranks it by urgency using NLP classification on message content and sender metadata
- Designed and built an incident reporting dashboard with Grafana and MongoDB used by 3 engineering teams

***Portfolio Analytics Engineering Intern*** **New York, NY**

*Swiss Re Insurance-Linked Investment Advisors* *May 2023 - August 2023*

- Built Python and SQL data pipelines to ingest and transform catastrophe bond exposure data, powering an exposure-monitoring tool used by the portfolio team

***Computer Science Teaching Assistant*** **Remote**

*Girls Who Code - Summer Immersion Program* *June 2022 - August 2022*

- Taught HTML, CSS and JavaScript to 120+ high school students and mentored an advisory group of 12

## PROJECTS

***Canvas Chart Maker: Photo-to-Needlepoint Chart Generator | React, TypeScript, Vite, ONNX Runtime Web, Vitest***

- Built a React and TypeScript web app that turns any photo into a stitchable needlepoint chart, used to design canvases for my needlepoint Etsy shop
- Implemented the image pipeline: k-means color quantization and CIEDE2000 color matching against the DMC floss palette, run in a Web Worker to keep the UI responsive
- Ran an ML background-removal model (BiRefNet) fully in the browser with ONNX Runtime Web on WebGPU, and covered the chart logic with Vitest unit tests

***Resell Photo Optimizer: AI Resale Listing Tool | React, TypeScript, Claude API, ONNX/WASM, Supabase***

- Built an AI pricing agent that reads clothing tags from photos, searches retail and resale listings, and suggests a price
- Automated product photos with ML background removal, centering, and exposure correction
- Generated ready-to-paste Depop listings from wear notes, with offline-first storage that syncs items across devices

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, Java, HTML/CSS
**Frameworks & Tools:** Next.js, React, oRPC, Drizzle ORM, PostgreSQL, Git, REST APIs, Vite, Vitest, MongoDB, Grafana
**Cloud & Infrastructure:** AWS (EC2), Docker, Kubernetes, OpenShift, Terraform, Ansible, Linux, CI/CD, Splunk
**ML & Data:** NLP, sentence embeddings (SBERT), TensorFlow, pandas, data pipelines, ONNX Runtime

## RESEARCH

***Undergraduate Research Assistant*** **Winston-Salem, NC**

*Wake Forest University - Economics Department* August 2024 - May 2025

- Built an NLP pipeline linking 50k+ patent descriptions to O*NET occupational tasks, using SBERT sentence embeddings and semantic similarity scoring to measure how inventions map to specific job functions